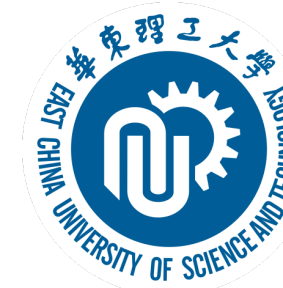


PVMark: Enabling Public Verifiability for LLM Watermarking Schemes



Private Detection. Public Verification. No Secret-Key Disclosure.

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PVMark enables anyone to verify the correct execution of LLM watermark detection while keeping the detector's secret key private.

Why Do We Need PVMark?

1. Risks of LLM Misuse



Academic Misconduct

Undisclosed AI use can undermine academic integrity and fairness.



AI-Generated Fake News

Fabricated content can mislead the public and damage trust.

2. Probability-Shift Watermarking

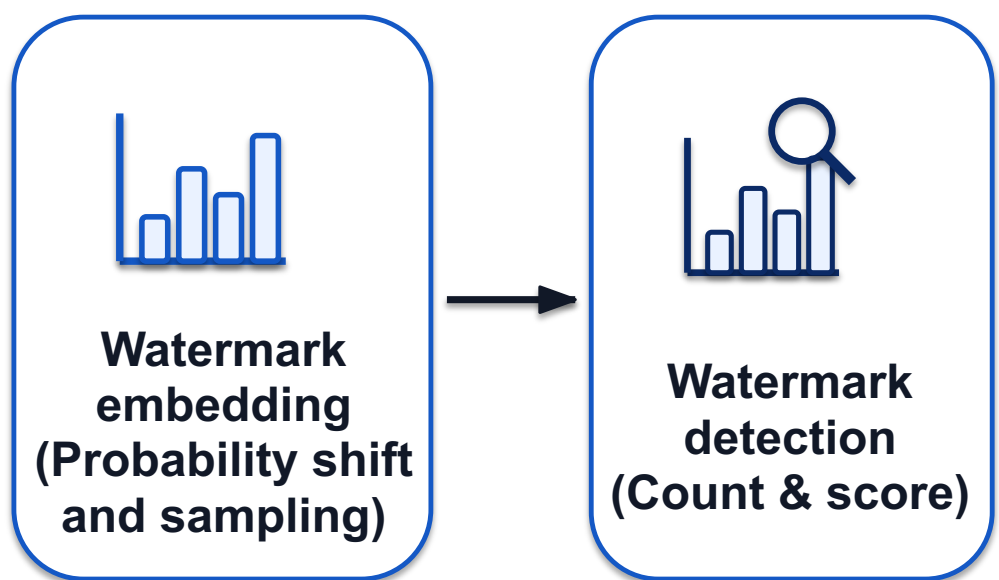
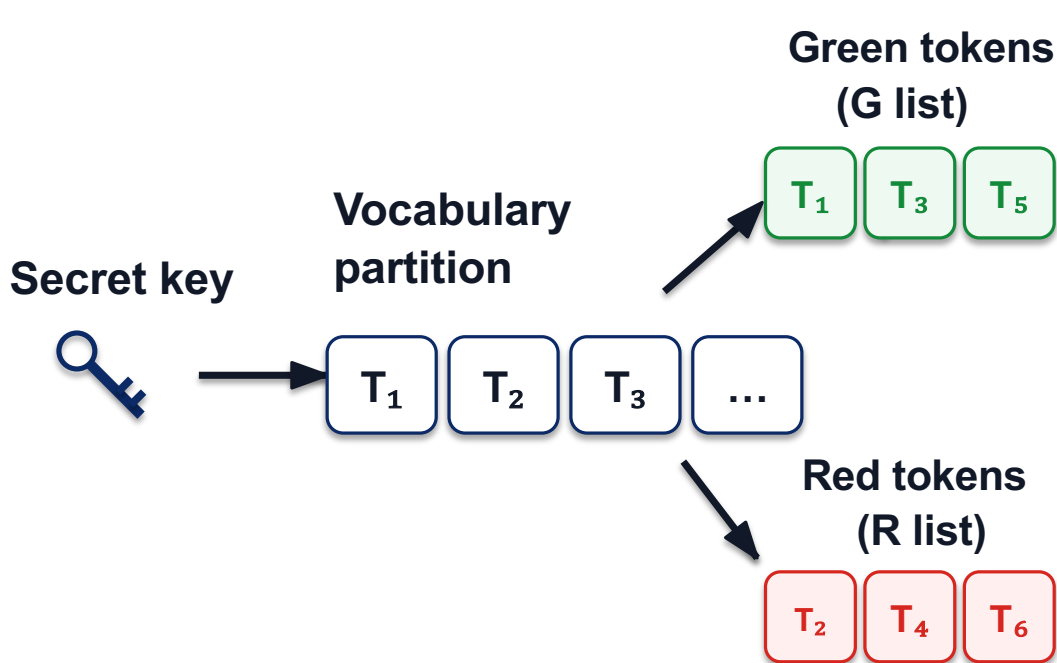
Representative schemes: KGW, SynthID-Text, Segment-Watermark.

Watermark Generation

- Random seed generation
- Vocabulary partitioning
- Probability shifting and sampling

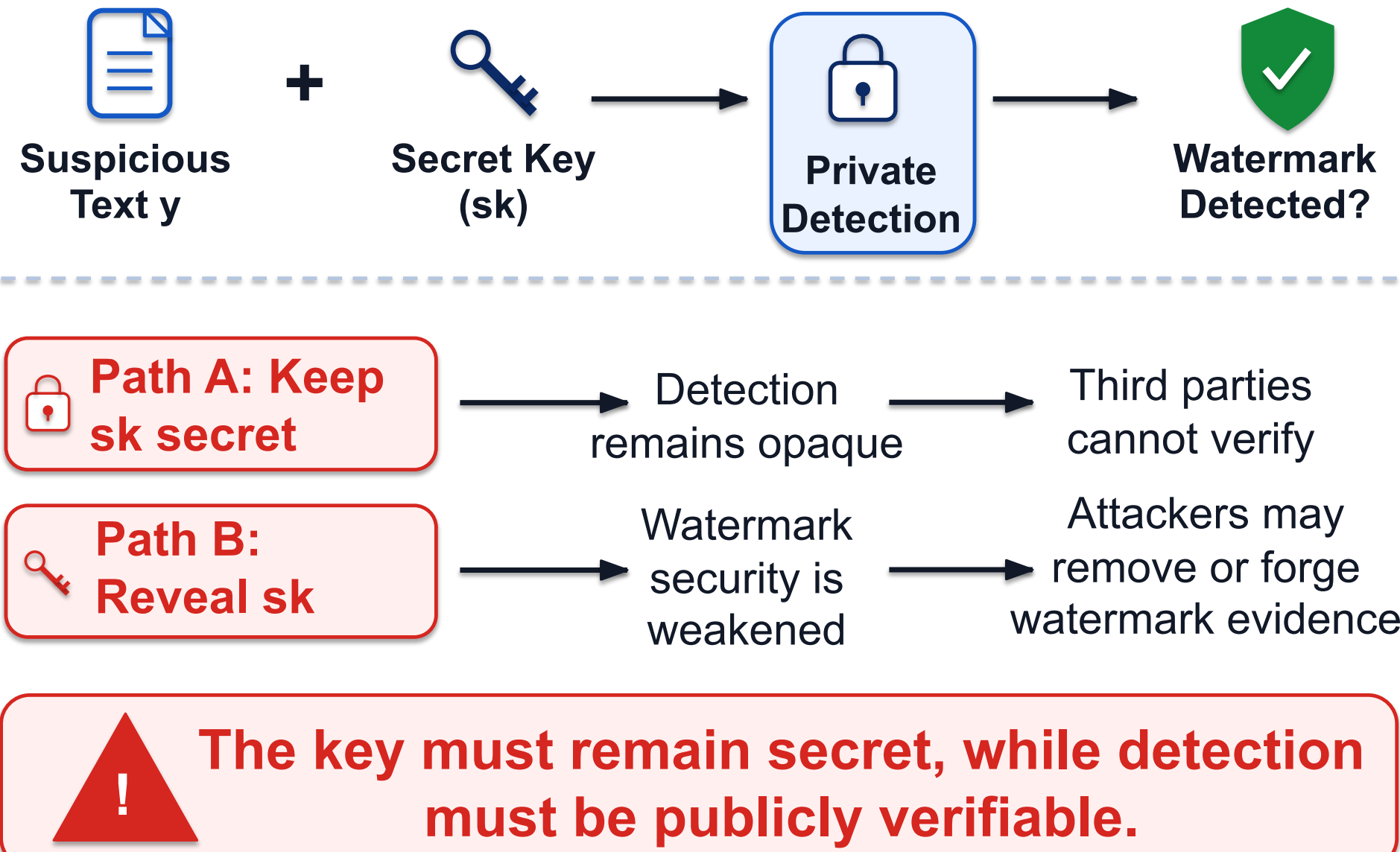
Watermark Detection

- Compute random seed
- Count green tokens
- Compute watermark score / message

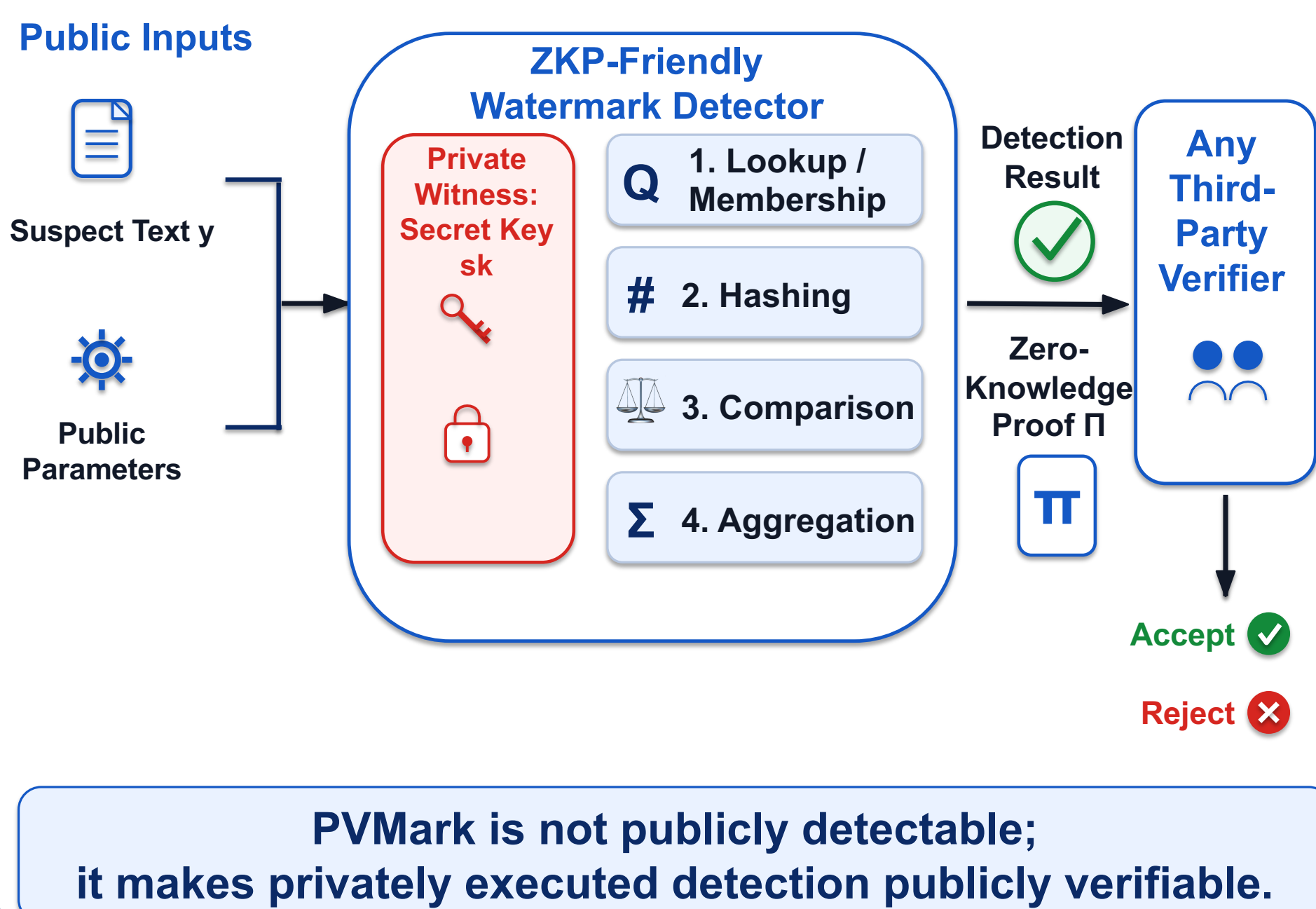


Embedding and detection are coupled through the same secret key.

Watermark Detection Creates a Trust Dilemma



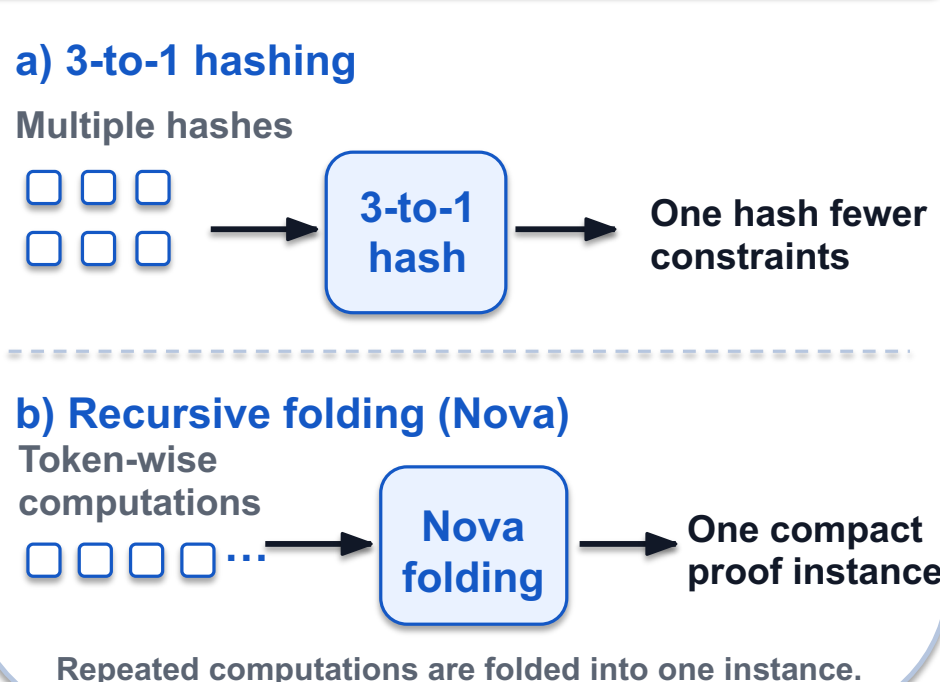
PVMark: Private Detection, Public Verification



Challenge 1: Make Detection Provable

Original	PVMark realization
f PRF	$\diamond$ MiMC / Poseidon / Poseidon2
Vocabulary partition	$\#$ Hash-threshold comparison
Hidden mapping	$\hat{h}$ Merkle membership proof
Token counting	Σ Comparison + aggregation

Challenge 2: Efficient Proof Generation



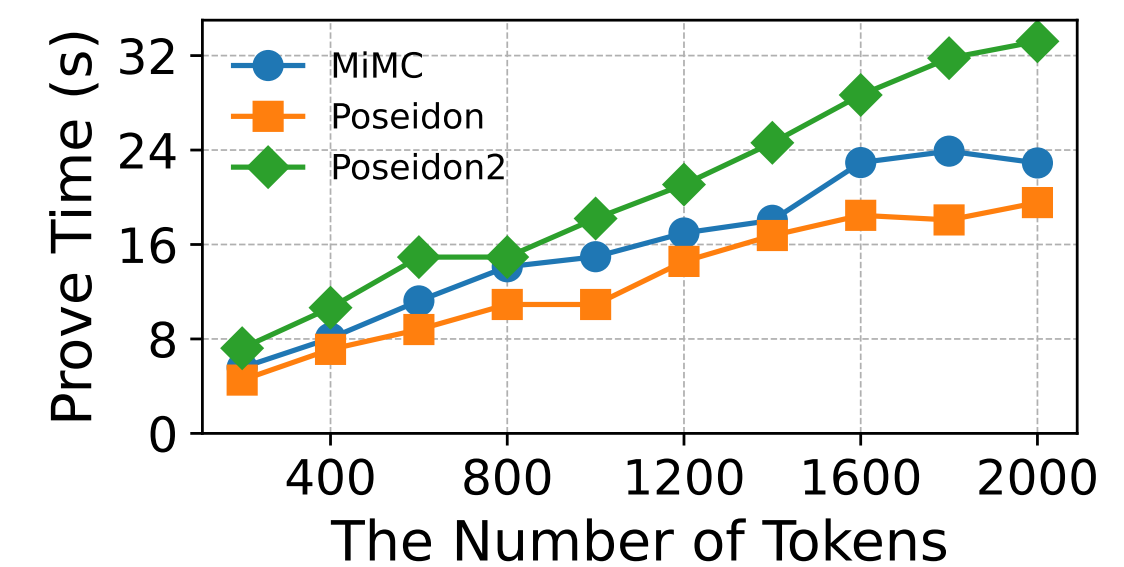
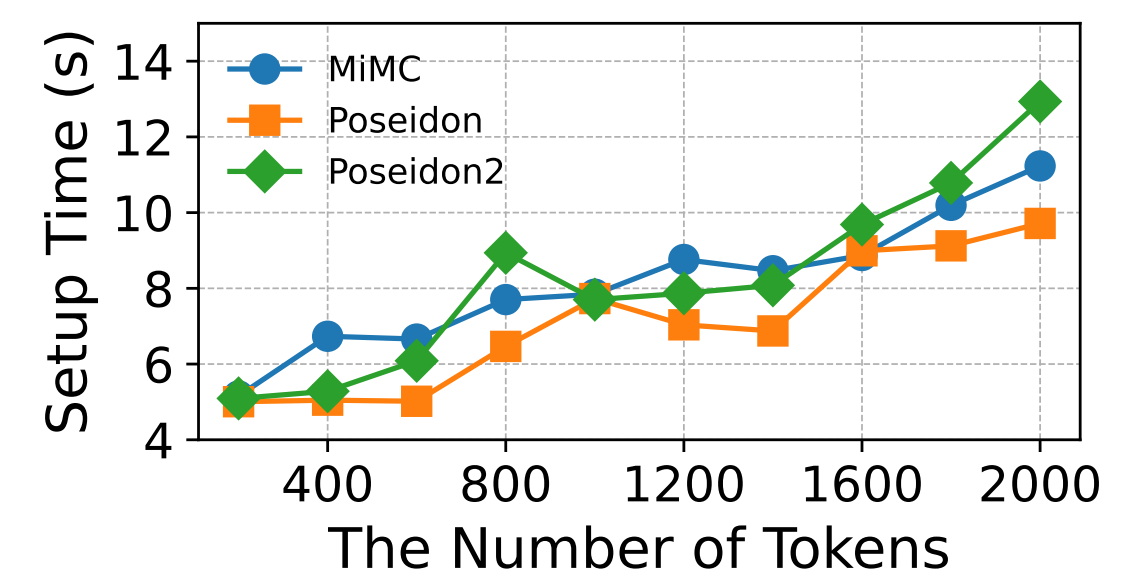
Experimental Highlights

Up to
99%
KGW Success Rate
OPT1.3B / C4 Dataset

Up to
100%
SynthID-Text Success Rate
GPT2 / ELI5 Dataset

< 2 s
Verification Time

Proof-Generation Cost (Setup time + Prove time)



2,000 tokens $\approx$ 30s

50,000 tokens $\approx$ 352s

Acceptable Proof Overhead

PVMark preserves effectiveness, fidelity, and robustness while adding public verifiability.

What PVMark Proves

- A specified watermark detector was correctly executed.
- Detection result corresponds to the suspect text.
- Verification succeeds without revealing the secret key.

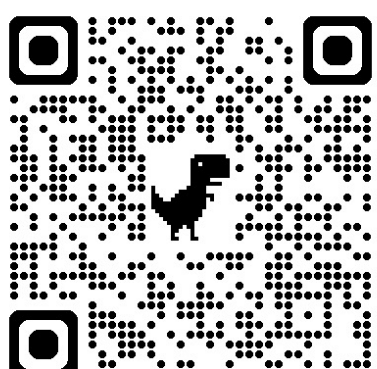
What PVMark Does Not Prove

- Authorship or intent
- Exclusive access to a model
- Robustness against every possible transformation

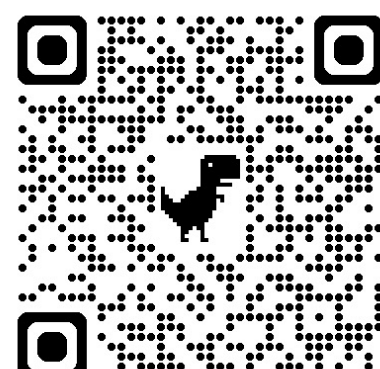
PVMark strengthens the credibility of watermark evidence for responsible AI and provenance tracing.



Paper:



Code / Artifact:



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